

MODUL UNIVERSITY VIENNA

SEID 2364: Societal and Ethical Impact of Data Science

SESSION 1

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We are going to take the education adventure together. We are going to marco, and go micro, and we are going to see why MU is the right place for you. Lets begin.

Opening Provocations

- Stream of consciousness.
- Capture your **immediate intuition**, not a polished or argued answer.
- Do not analyze or justify.
- Words or short phrases are enough.

(Write. Do not discuss yet.)



Opening
Provocations

What is “ethics”?

(Write. Do not discuss yet.)

Opening
Provocations

How do **YOU** currently
decide whether an action
is “right” or “wrong”?

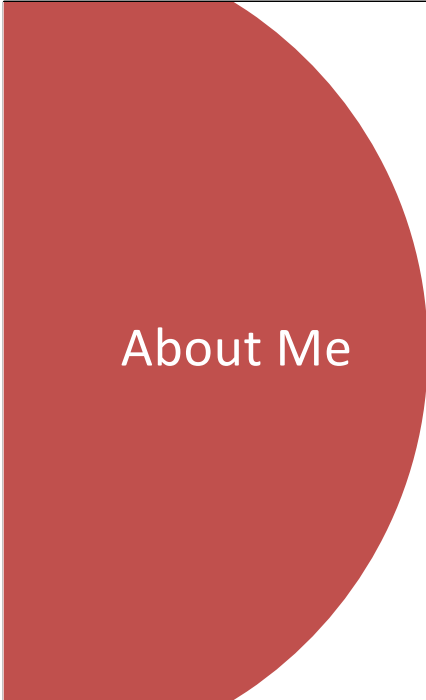
(Write. Do not discuss yet.)

Opening
Provocations

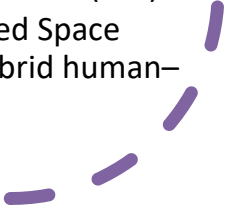
Can machines be
programmed to behave
“Ethically”?

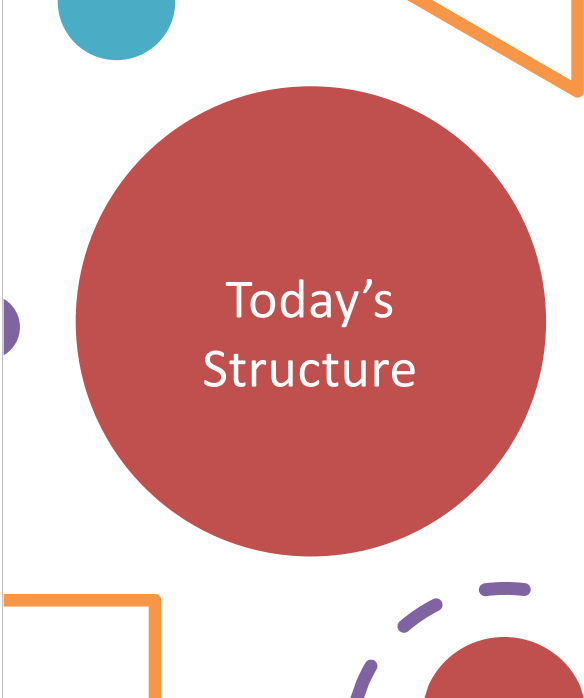
Why or why not?

(Write. Do not discuss yet.)



About Me

- Composer & Sound Designer (electronic and immersive media)
 - Early pioneer in Live Virtual Orchestra & networked performance systems
 - Long-term research in human-machine collaboration and distributed ensembles
 - Work integrating performers, real-time computation, AI systems, and spatial audio
 - Founder, Center for Holistic Integration (CHI)
 - Developer of the Balanced Blended Space (BBS) framework for analyzing hybrid human-machine systems
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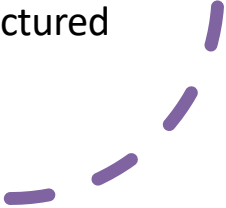


Today's Structure

- Course architecture
- Big ideas
- Wicked Problems
- Railroad switch experiment
- Moral Machine
- Reflection



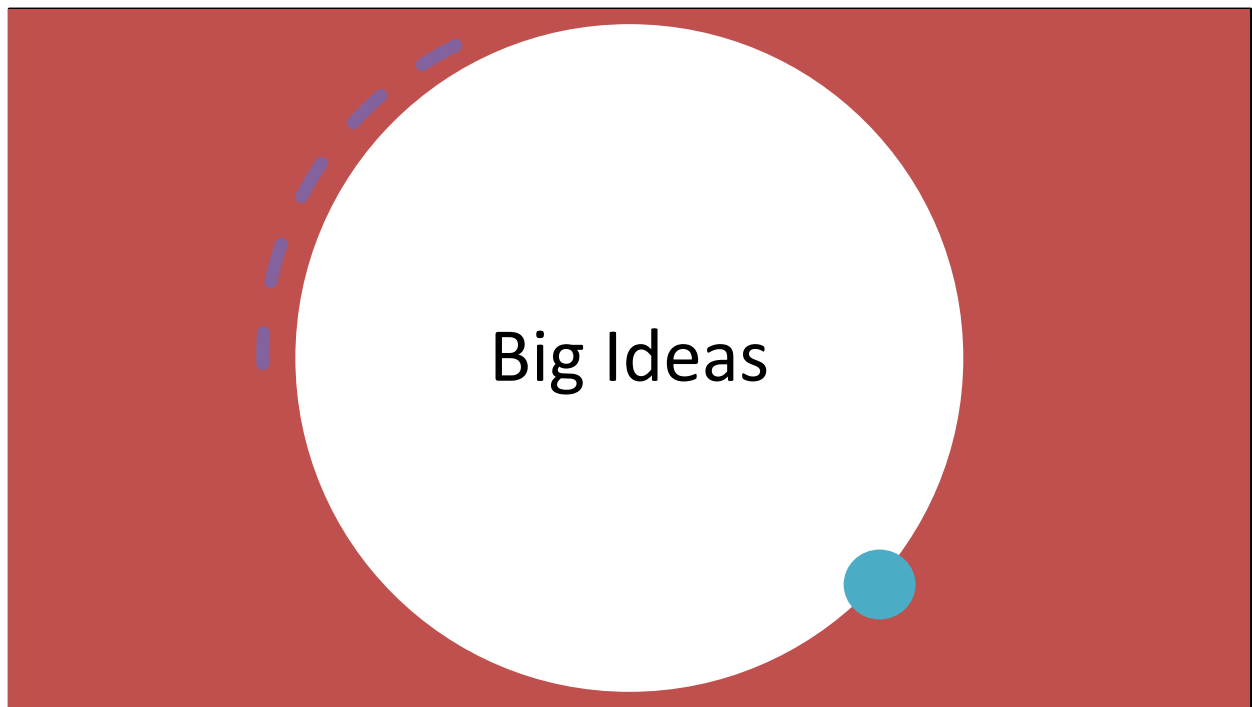
About This Course

- First offering in this format
 - Inquiry-driven, analytical, iterative
 - Ethics as practice in data-driven systems
 - Culminates in your own defensible ethical framework
 - Course evolves through structured exploration
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Course Platforms



- GitHub – version control & structured collaboration
- Zotero – shared research infrastructure
- Moodle – institutional interface
- OneDrive / Google – drafting & collaboration





Big Idea 1

Ethics Requires more than One.

Ethics emerges where actions affect other intelligences.
There is no ethics in a vacuum.

Ethics Requires More Than One Intelligence

Ethics arises in interaction. No interaction → no ethics. Ethics is about relationships between intelligences.



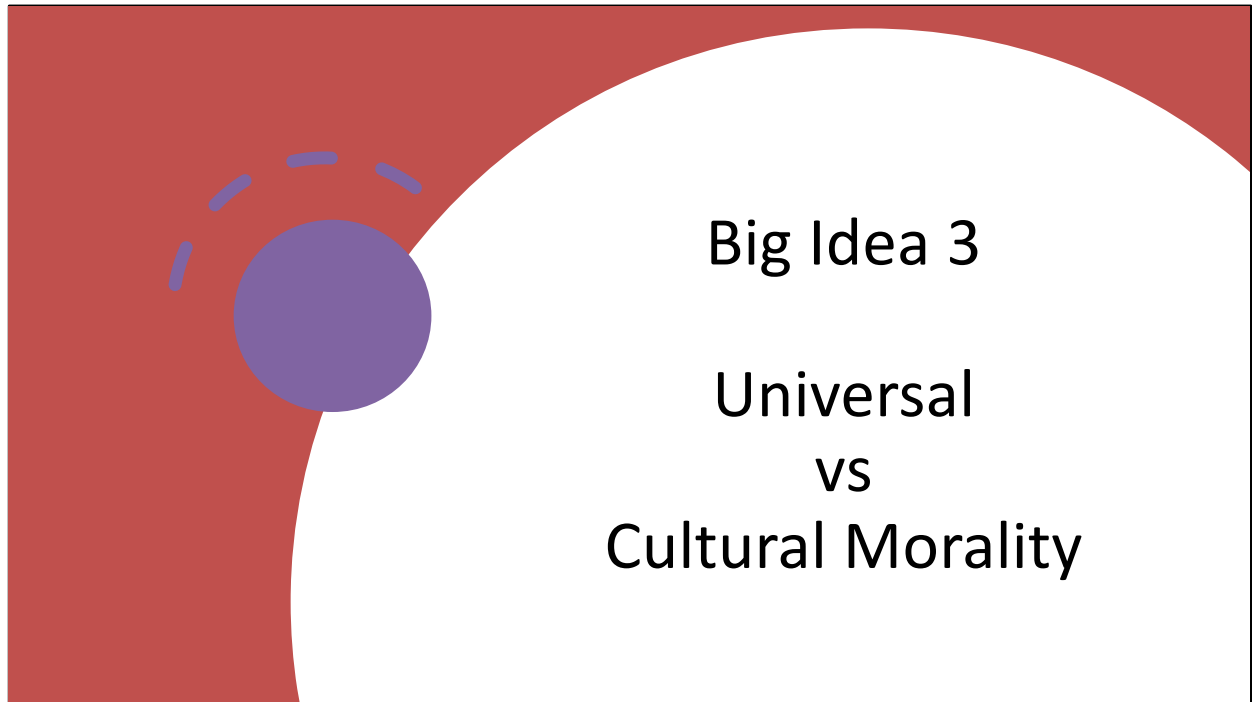
Ethics Requires Mediation

Define mediation.

Ethical action occurs through systems: language, law, technology, institutions, algorithms.

In data science, mediation is architectural.

System design, incentives, and mediation layers influence outcomes.
Architecture embeds values.



Are There Universal Ethical Principles?

Universal vs Cultural Morality

Are some principles absolute? Are others socially constructed?

If there are universal principles, what are the axioms?

Moral judgments shift across time and culture. Technological change reshapes moral boundaries.

What appears obvious in one era may be contested in another.

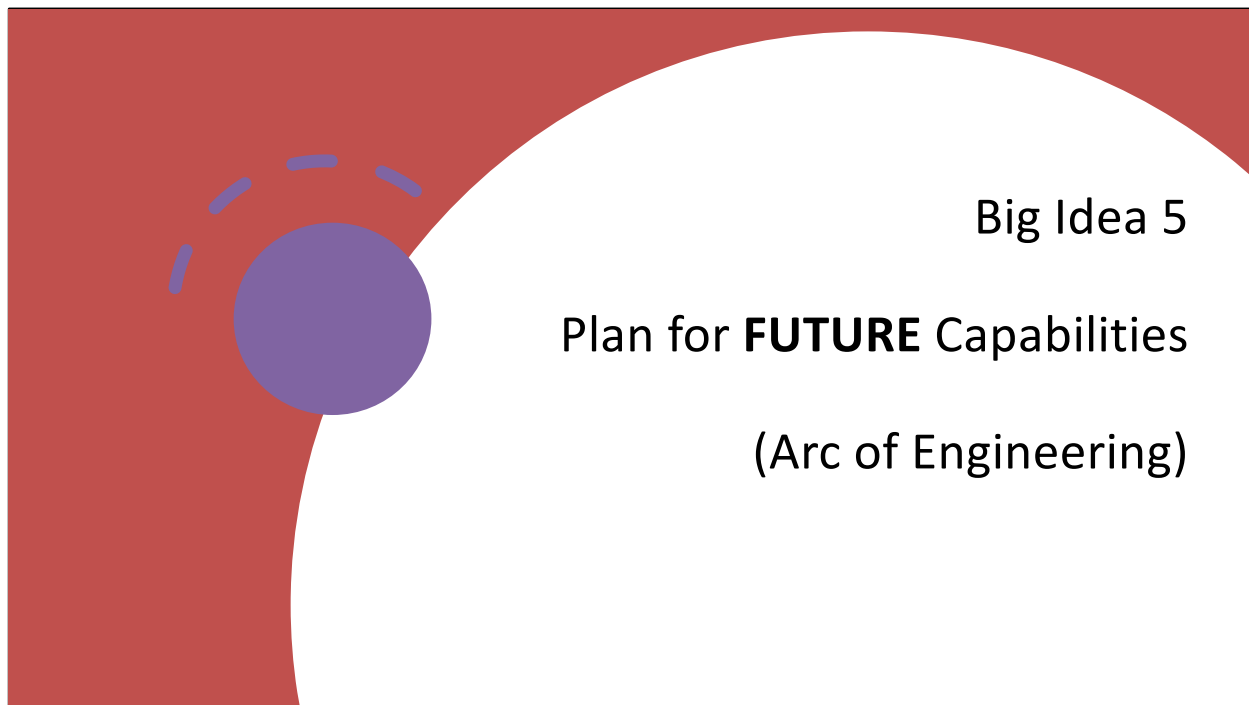


Intelligent Machines need Ethics too!

If machines exhibit intelligence and interact with society, they participate in ethical systems.

Interaction → Consequence. Consequence → Ethical relevance.
Therefore: Interactive systems require ethical architecture.
They classify. They allocate. They prioritize. They scale decisions.
In data science, you design this architecture.

As data scientists it is your responsibility to program ethical behavior.



We are living in a period of rapid — and accelerating — technological change.

Capabilities expand faster than regulatory or cultural adaptation.

Technologies begin centralized. They become distributed. They become embedded and ubiquitous.

When systems scale, their embedded ethical assumptions scale.

Some theorists describe this acceleration as approaching a technological "singularity" — a point where capability growth outpaces human comprehension and control.

Whether or not one accepts that term, we must anticipate systems whose impact exceeds present-day imagination.

Therefore: We must design ethical architectures not only for current systems, but for future capabilities that do not yet fully exist.

Engineering distributes judgment at scale.

When systems scale, assumptions scale.

Red Queen's Race

-
- "Well, in our country," said Alice, still panting a little, "you'd generally get to somewhere else—if you run very fast for a long time, as we've been doing."
 - "A slow sort of country!" said the Queen.
 - "Now, here, you see, it takes all the running you can do, to keep in the same place. If you want to get somewhere else, you must run at least twice as fast as that!" [🔗](#)





Big Idea 6

Context Changes Ethical Judgment

Context Changes Ethical Judgment

Even things that seem morally unambiguous may depend on context

The same behavior in different situations may be judged differently.

Intent, scale, power, and consequence reshape moral evaluation.

Ethics is not only about action, but about context.

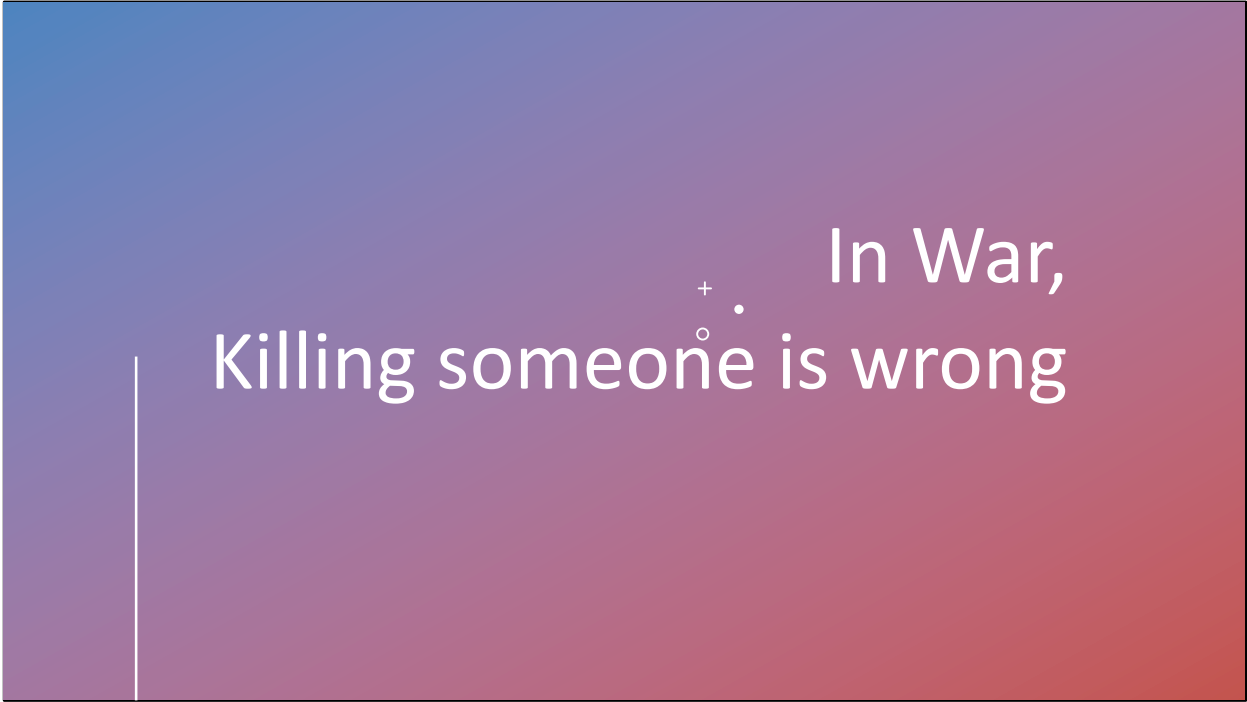
No definitive formulation.

Trade-offs are unavoidable.

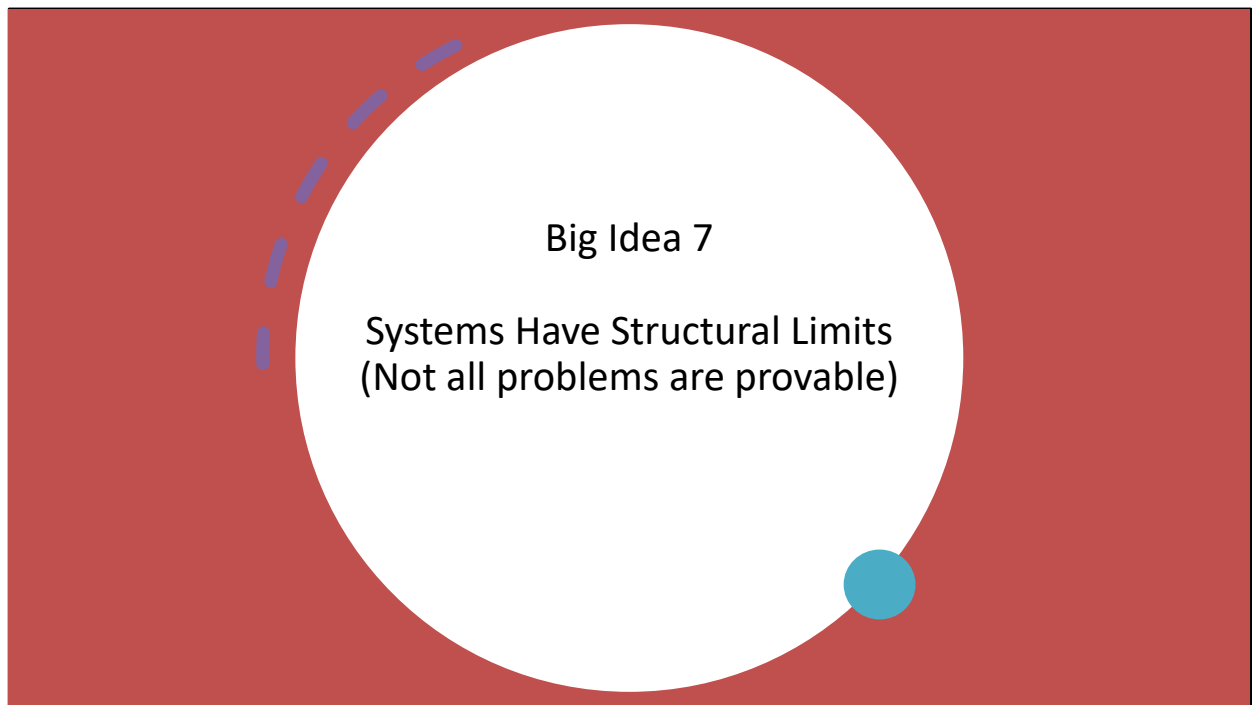
Solving one dimension may worsen another.

Example:

Killing someone⁺_° is wrong[•]



In War,
Killing someone is wrong



Systems have structural limits

No sufficiently expressive formal system resolves every conflict.
Ambiguity persists.

Some questions cannot be resolved within the system that generates them.
Gödel. Undecidability. The halting problem.
If a formal system is sufficiently expressive, there will be true statements it cannot prove.
There may be ethical questions that are structurally undecidable.

Example:

+
•
°
This statement is false

Worked example for Big Idea 7 (Systems Have Structural Limits).

This is the classic liar paradox. If the sentence is true, then it is false; if it is false, then it is true. The statement cannot be consistently assigned a truth value inside the system that produced it.

Use it the way we used "Killing someone is wrong / In war, killing someone is wrong" for Big Idea 6 — a deliberately tiny case that exposes a structural feature of the whole framework. Where slides 18 and 19 showed that context reshapes moral judgment, this example shows that some questions are undecidable from inside the system at all. Gödel, the halting problem, and self-reference all point to the same limit.

Why this matters for the rest of the lecture:

- It sets up Big Idea 8 and the wicked problems block (slides 22–25). Healthcare allocation and the Australian rabbits case are real-world analogues — no stopping rule, no clean reversibility, no single correct framing.
- It reframes the trolley sequence (slides 26–29) and the Moral Machine (slides 33–35). "You MUST choose" pressures students toward a clean rule, but the liar

paradox is a reminder that some dilemmas are structurally resistant to a clean rule.

- It directly anchors the Encoding Question on slide 31: would you encode your rule into a machine, and where does your rule fail? Undecidability is one place where it fails by construction.

- It connects back to Big Idea 2 (mediation is architectural) and Big Idea 4 (intelligent machines participate in ethical systems). If we hand contested decisions to algorithms, we are also handing them the system's structural limits — and those limits do not disappear because the output looks statistically objective.

Prompts to surface in discussion:

- Can you name an ethical question in data science that behaves like this sentence — true on one reading, false on another, with no stable answer inside the system?

- If a question is structurally undecidable, what is the responsible engineering move: refuse to decide, decide and disclose the residue, or escalate the decision outside the system?

- Tie back to the Red Queen's Race (slide 16): capabilities scale faster than our ability to resolve these limits. Designing for future capability (Big Idea 5) means designing assuming undecidability, not assuming it away.



Ethics Becomes Most Important When Problems Are Most Complex

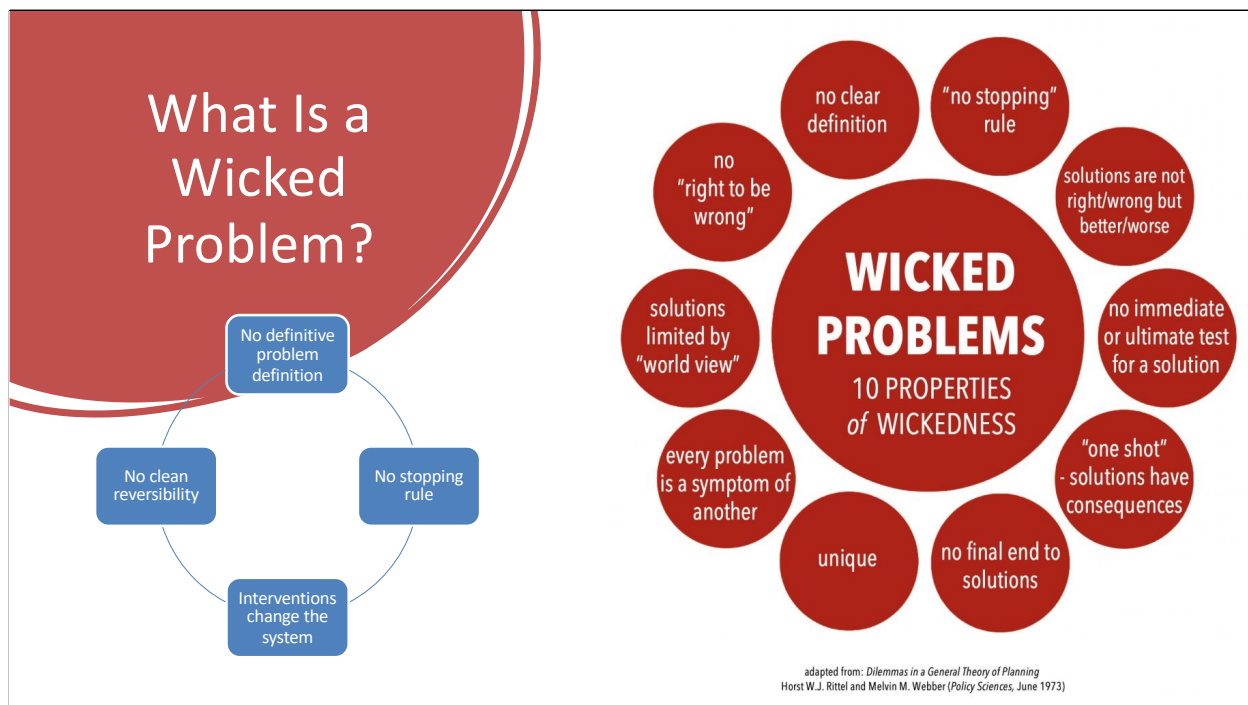
Ethical reasoning is easy when situations are simple. It becomes essential when:

- Trade-offs are unavoidable
- Stakeholders disagree
- Every option carries cost
- Outcomes are uncertain

In complex systems, what counts as the “correct” decision may depend on how the problem is framed.

Different approaches (utilitarian, rights-based, virtue-based, institutional, technical) can yield different conclusions.

These are often called "wicked problems."



No definitive problem definition (Stakeholders frame the problem differently; what is being optimized?)

No stopping rule (Solutions are better or worse — not true or false.)

Interventions change the system (Deployment alters incentives, behavior, and future states.)

No clean reversibility (Every trial counts; moral residue remains.)

Wicked Example A

Data-Driven Public Health Allocation

Scenario: A government deploys a predictive analytics system to allocate limited healthcare resources (e.g., ICU beds, cancer screening access, vaccination priority, organ transplants).

The system optimizes for measurable outcomes: • Survival probability • Cost-effectiveness • Expected life-years saved • Population-level efficiency

But alternative framings exist:

Optimization vs Justice Is the goal to maximize total lives saved? Or to correct historical disadvantage? Should disadvantaged populations receive weighted priority?

Efficiency vs Equity An efficiency model may allocate resources to patients most likely to recover quickly. An equity model may prioritize those historically underserved. These may produce opposite allocation decisions.

Historical Bias in Training Data If past healthcare access was unequal, training data encodes structural injustice.

A model trained on biased history may perpetuate or amplify disparities while appearing statistically objective.

Framing Question for Class: What is the system optimizing? Who decided that objective function? What would have to change in the framing to reverse the allocation outcome?



Wicked Example B: Rabbits in Australia

- Ecological Intervention: Rabbits in Australia
- Human introduction
- Cascading ecological damage
- Corrective intervention → new consequences
- How far ahead must prediction extend?

Scenario: European rabbits were introduced into Australia in the 19th century for hunting. With no natural predators and favorable conditions, populations exploded.

Consequences included: • Massive vegetation loss • Soil erosion • Native species displacement • Agricultural damage

The "solution" attempts included: • Hunting campaigns • Fencing projects •

Introduction of biological controls (e.g., myxomatosis virus, calicivirus)

Each intervention altered the ecological system. New unintended consequences emerged.

Ethical Tensions:

Human Responsibility Humans caused the imbalance. Do humans now have a moral obligation to intervene aggressively?

Intervention vs Non-Intervention Is it more ethical to act and risk new harms, or to refrain and allow damage to continue?

Prediction Limits Ecological systems are complex and nonlinear. Interventions may have delayed or cascading effects. No immediate test confirms long-term success.

Data Science Parallel: Predictive models can simulate outcomes, but model assumptions constrain foresight. Optimization targets may overlook ecological

or cultural variables.

Framing Question for Class: When systems are complex and adaptive, how far into the future must ethical responsibility extend? Who bears responsibility for second- and third-order effects?

Cross-Species disease transfer Diseases are known to jump species. Did scientists evaluate the probability that Mixomitosis could jump to humans or other organisms?

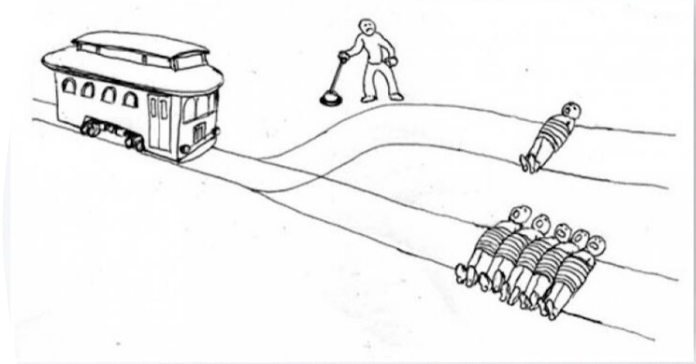
Simplifying the System

- **The Trolley Problem**

A trolley is heading toward five people.

You can pull a lever to divert it onto another track, where **one person** will be killed.

You **MUST** Choose

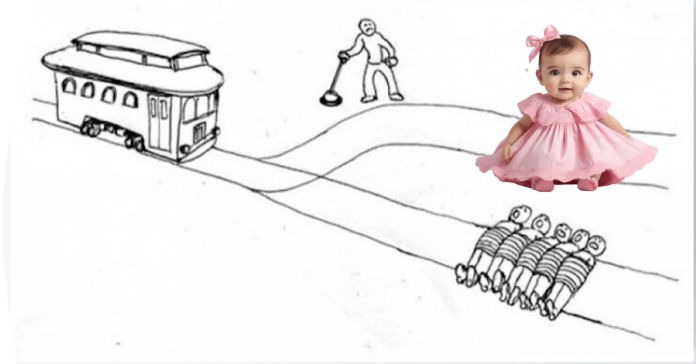


Escalation Value:

- **The Railroad Switch Problem**
A trolley is heading toward five people.

You can pull a lever to divert it onto another track, where **one baby** will be killed.

You **MUST** Choose

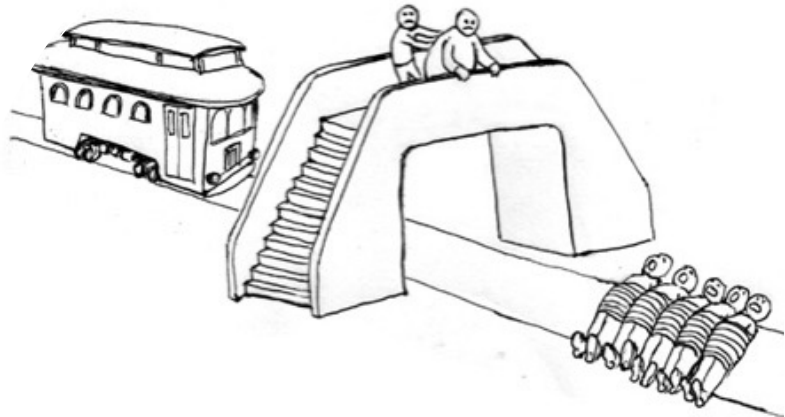


Escalation Effort :

- **The Railroad Switch Problem**
A trolley is heading toward five people.

You can push someone onto the tracks to prevent the trolley from hitting the people, but they will die.

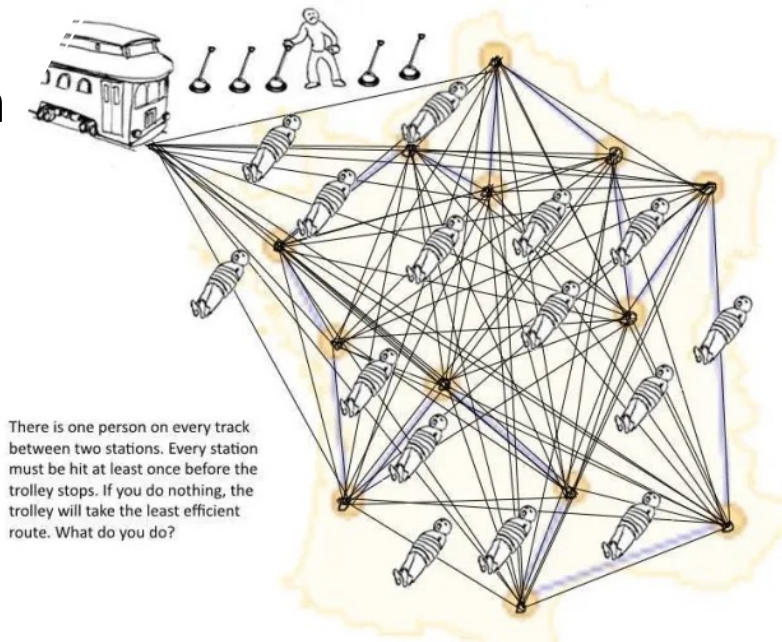
You **MUST** Choose



Combinatoric Trolley Problem

There is one person on every track between two stations. Every station must be hit at least once before the trolley stops. If you do nothing, the trolley will take the least efficient route.

What do you do?
You **MUST** Choose



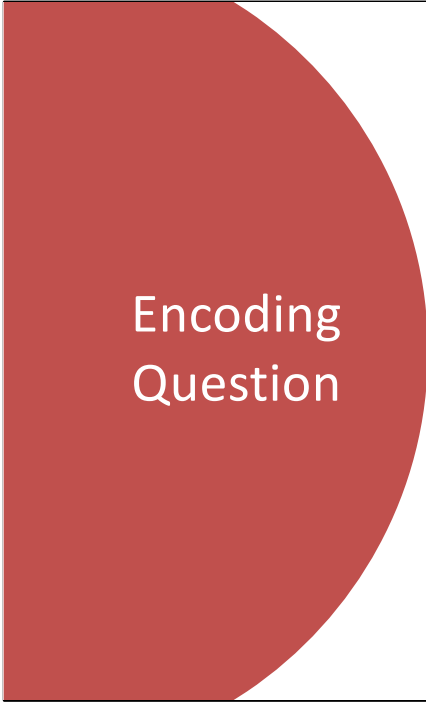


Did your thinking change?

Can you think of another situation that may cause you to change your decision?

What Rule did you use to make your decision?

Might there be other aspects that would contribute to this?



Encoding Question

- Would you encode your rule into a machine?
- Why or why not?
- Where does your rule fail?



Hybrid Human–Machine Systems

- Real-world decisions occur in distributed systems:
 - Human judgment
 - Technical architecture
 - Institutional constraints
-
- Responsibility is distributed.

Moral Machine

<https://www.moralmachine.net/> + •
○

Moral Machine Framing

Dataset of human responses

Reflects preferences, not moral authority

Raises questions about aggregation and governance

Does majority equal legitimacy?

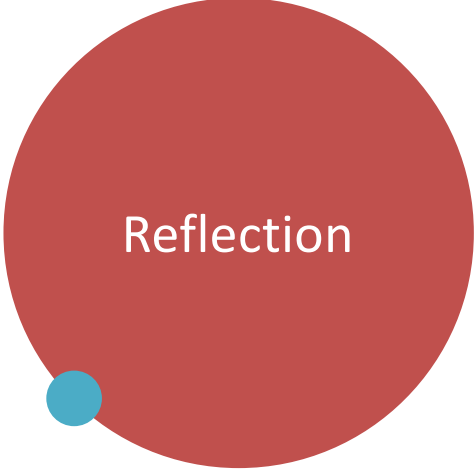
Statistical Morality?

If millions agree,

Is it correct?

If they disagree,

Who decides?



Reflection

- Return to your original definition.
- Has it changed?
- Can ethics be programmed?



Capability Orientation

- We are not only analyzing current computational systems.
- We are preparing to reason about future capabilities that may exceed today's frameworks.

Homework

- Account Preparation (GitHub + Zotero)
Journal Entry 1
Moral Machine
Royal Society Article
Dryad Dataset & Code
- <https://moodle.modul.ac.at/course/view.php?id=8146>

Closing Position



ETHICAL CLARITY IS
HARDER THAN IT
APPEARS.



DATA SCIENCE DOES
NOT ELIMINATE
MORAL AMBIGUITY.



COMPLEX SYSTEMS
GENERATE TRADE-
OFFS.



THE SEMESTER WILL
UNPACK THESE
TENSIONS.



Exit Reflection

- • What was your **Big Takeaway** today?
- What remains unclear or confusing?
- Which Big Idea needs deeper examination?
- What question should we explore next?